

DevOps Engineer

Step 2: Master Git & Scripting

CareerCompass • Detailed Concept Notes • Research-backed learning guide

Concept & Learning Objective

Infrastructure is code too. Git provides history, collaboration and review; scripting turns repeatable operational work into deterministic procedures. The important mindset is to prefer small, reviewable, reversible automation over manual production changes.

Git concepts include commits, branches, remotes, merges, pull requests, tags and conflict resolution. Rebase can create a cleaner local history but should be used carefully on shared branches. Shell scripts should validate inputs, handle failures and return meaningful exit codes.

Core Concepts — Detailed Breakdown

- Git object model: commits, trees and references at a conceptual level.
- Branches: isolated lines of development; merge integrates histories.
- Pull requests: review, discussion, automated checks and controlled integration.
- Rebase: replay commits onto another base; useful locally, risky when rewriting shared history.
- Tags/releases: identify deployable versions.
- Shell scripting: variables, quoting, conditions, loops, functions and command substitution.
- Exit codes: 0 usually means success; non-zero indicates failure.
- Idempotence: running a script twice should not create an unintended second copy of the desired state.
- Logging: print enough context to diagnose failures without leaking secrets.

What You Should Be Able To Do

- Create a feature branch and pull request.
- Resolve a merge conflict.
- Write a Bash script with validation and exit codes.
- Make a script safe to rerun.
- Store scripts and documentation in Git.

Hands-on Project / Practice

- Create a deployment-helper script that checks required environment variables, validates a target, builds an artifact and reports failure clearly. Add a dry-run mode.

Recommended References

- Git Documentation — <https://git-scm.com/doc>
- GitHub Documentation — <https://docs.github.com/>
- Git and GitHub Tutorial — <https://www.youtube.com/watch?v=AB3J8ufDYHQ>

Note: These notes are designed to teach the concepts and provide a practical learning path. Always prefer the linked official documentation for current APIs, SDK versions and platform-specific release requirements.